

Chains can't loop

```
system
You are the barista at Beanline Coffee. Maya wants a latte; find a milk that's actually in
stock.
```

```
human
Maya wants an oat-milk latte.
```

```
ai
tool_call: propose_milk(milk='oat milk')
```

```
tool
oat milk: out of stock
```

```
human
oat milk is out -- please propose another milk.
```

```
ai
tool_call: propose_milk(milk='almond milk')
```

```
tool
almond milk: out of stock
```

```
human
almond milk is out -- please propose another milk.
```

lap 3: re-spliced by hand

```
# context re-spliced by hand
# after every missed lap:
context = [
    system, human0,
    ai1, tool1, human1,
    ai2, tool2, human2,
]
ai3 = model.invoke(context)
# -> whole milk: in stock
```

Lap 3's context has grown to 8 messages, hand-appended lap by lap. That splice IS the loop.